

Owner-conditioned physical response, mixed-Gram information, shell-decay boundary, and stratified transversality

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-130 fixed the post-recombined terminal coordinate, the order “physical response first, predictable pull-back second”, the pointwise six-row Gram envelopes, the sharp deterministic balanced bridge, and a direct-low candidate. It did not construct the owner-complete production response or the shell constants required by T-050. R-131 tests that frontier from four independent directions.

1. It writes the exact finite-cylinder Hessian response after all owners have been assembled and proves the orthogonal coanalysis identity $R^*F = H_{\text{endpoint}}$. This is a conditional factorization theorem: an owner-complete physical form is an input, not a result of the matrix fixture.
2. It extracts a floor-uniform H^2 form bound for the deterministic current-square component, with exact coefficient $46959/(2000P)$ before the Sobolev embedding constant.
3. It proves that diagonal sample Grams and bounded pointwise multipliers do not determine the mixed conditional response or the required dyadic shell decay. These are method no-go results, not counterexamples to the production theorem.
4. It derives the exact two-channel and low-coupled acceptance formulas, an active-stratum lower bound for the radial coefficient Gram of Ξ , the common-phase full-tangent identification no-go, and a fixed-heat non-uniformity theorem. These identify a viable successor route while showing why neither Ξ nor a fixed heat alone supplies a global gap.

The result is T4 at fixed finite cutoff and positive floor. It proves no production C_{mix} , C_{far} , c_{bal} , R_0 , S_0 , d , k , matching-energy theorem, absolute anchor, strict augmented gap, $\text{OVERLAP}_{\text{src}}$, Nelson estimate, removal theorem, interacting measure, Sector-A closure, or T5–T7 promotion.

2. One-use owner ledger

The R-123 action owner, after all current, trace, heat, forest, rational, Cartan, and low coordinates have been assembled, is

$$\mathcal{A}_J(h) = \sum_{k,C} \mathbb{E} \mathcal{P}_{k,C}^{\text{comp}} + \frac{9}{20} \mathbb{E} \sum_k \|h_k\|^2 + \frac{3}{20} \mathbb{E} \|Z_K\|_6^6, \quad (2.1)$$

$$\mathcal{P}_{\text{comp}} = \frac{1}{2} \|\Phi\|^2 - \frac{1}{2} \Theta. \quad (2.2)$$

The source coefficient 9/20 therefore produces the Hessian coefficient 9/10. The terminal sextic coefficient 3/20 occurs once after terminal aggregation. Heat is internal to $P_\Sigma(C^T C)$, and the R-063 forest is an alternate expansion of the same trace owner. Neither is an additional positive reserve.

For the A1 Class-II coefficients, put

$$P = M_X^2 + \varepsilon_m = 4 + 10^{-12}, \quad (2.3)$$

$$a = \frac{c_{JJ}\alpha_X^2}{P}, \quad b = \frac{c_{JK}\alpha_X\beta_X}{P}, \quad c = \frac{c_{KK}\beta_X^2}{P}. \quad (2.4)$$

Completing the square from the manifest inputs, rather than pasting derived constants, gives

$$\alpha = \frac{c}{b+c} = \frac{5}{9}, \quad c_1 = \frac{(b+c)^2}{c} = \frac{243}{8000P}, \quad c_0 = a - \frac{b^2}{c} = \frac{3}{250P}. \quad (2.5)$$

Thus

$$\beta_{\text{op}} = 4(c_0 + c_1) = \frac{339}{2000P}. \quad (2.6)$$

There are two distinct Cartan quantities. The controlled regular FAR square has coefficient

$$4\alpha^2 c_1 = \frac{3}{80P}, \quad (2.7)$$

and occurs once only on the R-103 regular no-revisit class. The live local rational curl is

$$-\frac{40}{729} + \frac{2720}{729} = \frac{2680}{729}, \quad (2.8)$$

so the oriented energy coefficient is

$$c_C = \frac{1}{2} \frac{2680}{729} c_1 = \frac{67}{1200P}, \quad (2.9)$$

and the doubled full-cross coefficient is 67/(600P). Equations (2.7) and (2.9) are not two copies of one owner.

3. Conditional finite-cylinder response theorem

Let z, w be cylindrical physical variations and let E_f be the fixed conditional expectation associated with the assembled packet. Differentiating (2.2) before any source pullback gives the complete physical bilinear form

$$\begin{aligned} \beta_{\text{comp}}(z, w) = \sum_{k, C} \mathbb{E}\{ \text{Re}\langle E_f \dot{J}_z, E_f \dot{J}_w \rangle + \text{Re}\langle \Phi, E_f \ddot{J}_{z, w} \rangle \\ - \frac{1}{2} D^2 \Theta[z, w] \}. \end{aligned} \quad (3.1)$$

Let Q_{comp} be its physical Riesz representative, let Q_6 be the Hessian of the once-owned terminal sextic, and let L_{pred} be the physical synthesis of a predictable source. Then

$$D^2 \mathcal{A}_J(h) = \frac{9}{10} I + L_{\text{pred}}^* (Q_{\text{comp}} + Q_6) L_{\text{pred}}. \quad (3.2)$$

The source term is outside the physical pullback. If $v \in \text{Ker } L_{\text{pred}}$, the second term in (3.2) vanishes while

$$\langle v, (9/10) I v \rangle = \frac{9}{10} \|v\|^2. \quad (3.3)$$

Let $\mathfrak{C}x = (\Pi_m x)_m$ be orthogonal shell coanalysis, and define

$$R = \mathfrak{C}L_{\text{pred}}, \quad F = \mathfrak{C}(Q_{\text{comp}} + Q_6)L_{\text{pred}}. \quad (3.4)$$

Since $\sum_m \Pi_m = I$ on the finite cylinder,

$$R^*F = L_{\text{pred}}^* \left(\sum_m \Pi_m \right) (Q_{\text{comp}} + Q_6)L_{\text{pred}} = H_{\text{endpoint}}. \quad (3.5)$$

The reverse expression is the adjoint of the same owner, not a second reserve. Moreover

$$\|F\|_{\ell_m^2}^2 = \|(Q_{\text{comp}} + Q_6)L_{\text{pred}}\|^2. \quad (3.6)$$

Thus orthogonal shell analysis preserves the full response norm; it creates no off-diagonal decay by itself.

The primary exact matrix fixture takes an arbitrary symmetric Q_{comp} , a positive Q_6 , and a synthesis with one vertical kernel. Writing $A = (Q_{\text{comp}} + Q_6)L_{\text{pred}}$, it checks (3.2)–(3.6), including $F^*F = A^*A$. The arbitrary fixture proves the factorization once an owner-complete form is supplied. It does not construct the production Q_{comp} or prove that it is uniformly bounded in cutoff, floor, chart, control, filtration, visit, or shell.

4. A floor-uniform deterministic current-square component

Consider only the deterministic current-square energy

$$E_{\text{cur}}(u) = \frac{1}{2} \int \sum_i \langle D_i u, B_e(u) D_i u \rangle \, dx. \quad (4.1)$$

R-130 proved the pointwise complete-Gram contraction envelopes

$$\|B_e(u)\| \leq \beta_{\text{op}}|u|^2, \quad \|DB_e(u)[z]\| \leq L_6|u||z|, \quad \frac{1}{2}|D^2(B_e : Q)(u)[z, z]| \leq H_6\|Q\|_{\text{op}}|z|^2 \quad (4.2)$$

with

$$L_6 = \frac{1143}{250P}, \quad H_6 = \frac{7083}{500P}. \quad (4.3)$$

These constants are imported from the independently verified R-130 authority. They are not reconstructed from hidden hardcoded frame numbers.

Differentiating (4.1) twice in a diagonal direction gives the three exact multiplicities

$$|D^2 E_{\text{cur}}(u)[z, z]| \leq \int \{ \beta_{\text{op}}|u|^2|Dz|^2 + 2L_6|u||z||Du||Dz| + H_6|z|^2|Du|^2 \} \, dx. \quad (4.4)$$

The $H^2(\mathbb{T}^3) \hookrightarrow L^\infty$ embedding and Cauchy give

$$|D^2 E_{\text{cur}}(u)[z, w]| \leq C_{H^2 \rightarrow L^\infty}^2 K_{\text{diag}} \|u\|_{H^2}^2 \|z\|_{H^2} \|w\|_{H^2}, \quad (4.5)$$

where polarization is used for z, w and

$$K_{\text{diag}} = \beta_{\text{op}} + 2L_6 + H_6 = \frac{46959}{2000P}. \quad (4.6)$$

This is a genuine floor-uniform component theorem. It includes neither the conditional mixed Gram, the trace, heat, forest, rational, Cartan, low, nor chart-response derivatives. Consequently (4.5) is not the production C_{mix} or C_{far} theorem.

5. Diagonal Gram data do not determine the mixed response

For a random frame C_f and payload V_f , the conditional mean square is the replica expression

$$\|\mathbb{E}_f C_f V_f\|^2 = \mathbb{E}_{f,f'} \langle V_f, C_f^* C_{f'} V_{f'} \rangle. \quad (5.1)$$

It depends on the mixed Gram $C_f^* C_{f'}$, not only on the diagonal sample Gram $C_f^* C_f$.

On a two-point probability space set

$$C_+(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}, \quad C_-(t) = C_+(-t). \quad (5.2)$$

Every sample satisfies

$$C_\pm(t)^* C_\pm(t) = I, \quad (5.3)$$

so all derivatives of the diagonal Gram vanish. But

$$\mathbb{E} C_f(t) = \cos t I, \quad \|\mathbb{E} C_f(t) v\|^2 = \cos^2 t \|v\|^2, \quad (5.4)$$

and hence, for $\|v\| = 1$,

$$\frac{d^2}{dt^2} \|\mathbb{E} C_f(t) v\|^2 \big|_{t=0} = -2. \quad (5.5)$$

The independent verifier repeats the separation with rational rotations having entries $4/5$ and $3/5$. Therefore no argument using only the R-130 diagonal Gram constants can determine the mixed conditional response. A production proof must retain the full conditional replica Gram, including trace, heat, forest, rational, Cartan, and paid owners before absolute values.

This is the formal no-go NG-2026-07-31-A13-DIAGONAL-GRAM-TO-MIXED-CONDITIONAL-RESPONSE. It refutes an inference from incomplete data, not the desired production estimate itself.

6. Bounded multipliers do not imply dyadic shell decay

Let $r < m$, set $n_r = 2^r$, $n_m = 2^m$, and put

$$Q_{m,r}(x) = \cos((n_m - n_r)x), \quad \|Q_{m,r}\|_\infty = 1. \quad (6.1)$$

Here Π_j is the Laurent-coordinate projection onto the single mode $e^{in_j x}$ in this prototype. It is not an assertion about which boundary mode a particular smooth production Littlewood–Paley window assigns to shell j ; equivalently, the two modes may be moved into any chosen interior lattice coordinates of two separated windows. Multiplication of $e^{in_r x}$ by $Q_{m,r}$ has Fourier coefficient $1/2$ at n_m . Thus

$$\|\Pi_m M_{Q_{m,r}} \Pi_r\| \geq \frac{1}{2}. \quad (6.2)$$

If pointwise boundedness alone implied prototype estimates

$$\|\Pi_m M_Q \Pi_r\| \leq C_{\text{mix}} 2^{r-2m}, \quad \|\Pi_m M_Q \Pi_r\| \leq C_{\text{far}} 2^{r-4m}, \quad (6.3)$$

then (6.2) would force

$$C_{\text{mix}} \geq 2^{2m-r-1}, \quad C_{\text{far}} \geq 2^{4m-r-1}. \quad (6.4)$$

For the executed test $r = 3, m = 19$, these are 2^{34} and 2^{72} . They diverge with m . The selected shell coefficient $1/2$ must not be confused with the multiplier supremum 1.

Frozen coefficient derivatives also miss state dependence. With $B(u) = u^2$ and $Q_n(u) = \sin(2\pi n u)$,

$$\frac{d}{du} \{u^2 Q_n(u)\} \big|_{u=1} = 2\pi n, \quad (6.5)$$

while the frozen- Q B derivatives are unchanged. The independent route uses a bounded rational alternative to reach the same conclusion. Thus L_6, H_6 alone do not control DQ or the shell variation of a complete state-dependent response.

This is the formal no-go NG-2026-07-31-A13-BOUNDED-MULTIPLIER-TO-SHELL-DECAY. A production-specific paraprodut, commutator, or complete-Gram cancellation theorem may still supply the missing decay.

7. Exact conditional acceptance simplex

Let the remaining source and sextic diagonals after all once-owned charges be

$$e = 2\eta - r_0, \quad f = 2\zeta - s_0. \quad (7.1)$$

If a is the full cross norm after all owners have been assembled, and $e, f \geq 0$, the two-channel matrix is positive semidefinite when

$$a \leq 2\sqrt{ef}. \quad (7.2)$$

It is positive definite when $e, f > 0$ and the inequality is strict; equality in (7.2) gives zero lower margin. Its exact lower margin is

$$m_2 = \frac{e + f - \sqrt{(e - f)^2 + a^2}}{2}. \quad (7.3)$$

For a low block $D \succeq dI$ coupled with norm k , a strict sufficient condition is

$$k^2 < dm_2, \quad (7.4)$$

and the resulting three-channel margin is

$$\mu_3 = \frac{m_2 + d - \sqrt{(m_2 - d)^2 + 4k^2}}{2} > 0. \quad (7.5)$$

An affine absolute anchor additionally needs the Douglas range condition and the reduced-solution cost. Semidefiniteness alone is insufficient.

The complete cross majorant must have the convention

$$a \leq 2C_{\text{br}}|c_{\text{bal}}| + \ell \left[\sqrt{8/7} C_{\text{mix}} 2^{-j_0 - 2C} + \sqrt{224/223} C_{\text{far}} 2^{-3j_0 - 4C} \right] + a_{\text{residual}}, \quad (7.6)$$

where $\ell = 1$ only if the reported constants already include the legal source pullback. Otherwise the R-129 factor $\|L_{\text{pred}}\|$ is still required.

For the old zero-diagonal diagnostic, recompute the symbol constant exactly from A1:

$$c_{\text{sym}} = \frac{26000000000947494031}{239930905990000000000}, \quad A_0 = c_{\text{sym}}^{-1}, \quad C_{\text{br}} = \sqrt{32A_0}. \quad (7.7)$$

The rounded A8 JSON value is used only as a tolerance check. With

$$\eta_0 = \frac{197}{440} - 2c_0, \quad \zeta_0 = \frac{3}{25}, \quad K_0 = 4\sqrt{\eta_0\zeta_0}, \quad (7.8)$$

the local Cartan cross is

$$A_C^{\text{eff}} = 2C_{\text{br}}c_C = 0.4797278958\dots, \quad K_0 = 0.9209323339\dots \quad (7.9)$$

It occupies 0.5209154659... of that diagnostic ceiling and leaves 0.4412044381.... At collar $C = 1$ the remaining checksum is

$$0.2672612419 C_{\text{mix}} 2^{-j_0} + 0.06263997778 C_{\text{far}} 2^{-3j_0} \leq 0.4412044381. \quad (7.10)$$

This is a viability filter only. The production constants and all same-channel debts remain unknown, and the old source/sextic reserves cannot be spent once to complete a force and again to pay its Hessian.

8. Stratified Ξ transversality

Write

$$a = \operatorname{Re}(u^*v), \quad s = \operatorname{Re}(\bar{\chi}w), \quad h = u_1v_2 - u_2v_1, \quad (8.1)$$

$$r = |u|^2, \quad \rho = r + |\chi|^2, \quad d = \rho + \varepsilon_\rho, \quad \lambda = \frac{\alpha r}{d}. \quad (8.2)$$

Because $0 \leq \lambda \leq \alpha = 5/9 < 1$, the exact R-130 coordinate gives

$$\frac{1}{4}\|\Xi\|^2 = c_0a^2 + c_1\{(1-\lambda)a - \lambda s\}^2 + (c_0 + c_1)|h|^2. \quad (8.3)$$

The radial (a, s) Gram is

$$G_\lambda = \begin{pmatrix} c_0 + c_1(1-\lambda)^2 & -c_1\lambda(1-\lambda) \\ -c_1\lambda(1-\lambda) & c_1\lambda^2 \end{pmatrix}. \quad (8.4)$$

Its trace and determinant are

$$T_\lambda = c_0 + c_1\{(1-\lambda)^2 + \lambda^2\}, \quad \det G_\lambda = c_0c_1\lambda^2. \quad (8.5)$$

The eigenvalues are

$$\lambda_\pm = \frac{T_\lambda \pm \sqrt{T_\lambda^2 - 4c_0c_1\lambda^2}}{2}. \quad (8.6)$$

On $0 \leq \lambda \leq 1$, $T_\lambda \leq c_0 + c_1$. Since $G_\lambda \succeq 0$,

$$\boxed{\lambda_-(G_\lambda) = \frac{\det G_\lambda}{\lambda_+(G_\lambda)} \geq \frac{c_0c_1}{c_0 + c_1}\lambda^2.} \quad (8.7)$$

(8.7) is a quantitative lower bound for the radial coefficient pair (a, s) , and (8.3) separately retains the wedge channel $(c_0 + c_1)|h|^2$. It is not a coercive estimate for the full tangent norm under only the natural common phase-horizontal condition $\Im(u^*v + \bar{\chi}w) = 0$, which is real-metric orthogonality to $(iu, i\chi)$. Indeed, with $u = (1, 0)$, $\chi = 1$, $v = (i, 0)$, and $w = -i$, one has $a = s = h = 0$ while $r|v|^2 + |\chi|^2|w|^2 = 2$. Thus the tempting full-norm identification is false; only the explicitly displayed coefficient seminorm is controlled. This is recorded as NG-2026-07-31-A13-NATURAL-PHASE-HORIZONTAL-XI-METRIC-IDENTIFICATION. The radial spectral order itself is sharp because

$$\lim_{\lambda \downarrow 0} \frac{\lambda_-(G_\lambda)}{\lambda^2} = \frac{c_0c_1}{c_0 + c_1}. \quad (8.8)$$

At a pure singlet, the coefficient decays like r^2/d^2 . Hence neither (8.7) nor the wedge channel extends to a state-uniform positive metric gap for the full tangent space. The radial calculation sharpens the stratified coefficient geometry but does not revive the retired global metric-error-bound route.

9. Fixed heat cannot supply a uniform transverse gap

Take the background $(u, \chi) = (0, T)$ and the aligned singlet variation $(v, w) = (0, 1)$. Add a fixed heat sample (U, X) and put $R = |U|^2$. The relevant Ξ output is

$$4c_1\alpha^2R^2 \frac{\{\operatorname{Re}(T + X)\}^2}{\{R + |T + X|^2 + \varepsilon_\rho\}^2}. \quad (9.1)$$

For every fixed heat sample this tends to zero as $T \rightarrow \infty$. If $y = |T + X|^2$, then

$$\frac{R^2 y}{(R + y + \varepsilon_\rho)^2} \leq \frac{R^2}{4(R + \varepsilon_\rho)} \leq \frac{R}{4}, \quad (9.2)$$

because

$$(R + \varepsilon_\rho + y)^2 - 4(R + \varepsilon_\rho)y = (y - R - \varepsilon_\rho)^2. \quad (9.3)$$

Thus dominated convergence applies to any fixed heat with finite doublet second moment, and the averaged contribution also tends to zero. The executed 64-atom product Rademacher law has zero mean and identity covariance in all six real heat coordinates. Its four doublet signs give the constant radius $R = 4$ and integrate out, so (9.1) collapses to four distinct singlet values, each with multiplicity sixteen. Its response is positive at every finite T but has the exact asymptotic

$$\lim_{T \rightarrow \infty} T^2 H(T) = 4c_1 \alpha^2 R^2. \quad (9.4)$$

This proves **NG-2026-07-31-A13-FIXED-HEAT-UNIFORM-TRANSVERSALITY**. It does not exclude state-dependent or scale-adapted heat, nor a coupled source-sextic-response argument. Indeed the once-owned sextic has aligned ray Hessian

$$\frac{d^2}{dt^2} \frac{3}{20} (T + t)^6 \big|_{t=0} = \frac{9}{2} T^4. \quad (9.5)$$

This supplies large-amplitude curvature, but its complete interaction with the mixed response, low block, covariance defect, and absolute anchor has not been proved.

10. Proof-search evidence map

Every substantive route below is recorded in the append-only exploration ledger as EXP-000483–EXP-000495. The table is a compact view, not a replacement for the exact question, finite checks, evidence, boundary, and revisit condition in those records.

Route	Verdict	Boundary or next condition
Complete owner census	advanced	Use (2.1) once; no heat/forest/source or Cartan duplication.
Physical response then pullback	advanced conditionally	Production Q_{comp} and uniform norm still missing.
Deterministic current H^2 form	advanced	Component only; coefficient is $46959/(2000P)$.
Diagonal Gram controls mixed response	failed	Retain $C_f^* C_{f'}$.
Bounded multiplier gives shell decay	failed	Need production spatial regularity or cancellation.
Frozen- Q envelopes control DQ	failed	Differentiate the full response.
Two/three-channel acceptance	advanced conditionally	Supply $C_{\text{mix}}, C_{\text{far}}, c_{\text{bal}}, R_0, S_0, d, k$.
Local Cartan old-budget test	viable diagnostic	Not a global operator bound; no budget reuse.
Ξ radial coefficient-pair coercivity	advanced	Degenerates quadratically in λ ; wedge retained separately.
Common-phase quotient gives full-tangent coercivity	failed	A phase-invisible tangent has coefficient seminorm zero and weighted norm two.
Ξ gives a global gap	failed	Pure-singlet transition has quartic energy decay.
Fixed heat gives uniform fill	failed	Any fixed law vanishes on large singlet rays.
Sextic repairs the complement	inconclusive	Prove the complete coupled three-channel form and anchor.

The route pruning is material. Repeating diagonal-Gram differentiation, bounded-multiplier shelling, or fixed-heat transversality cannot close T-050. The live route is an owner-complete mixed-replica multiplier theorem with a state- and scale-adapted transverse complement, followed by the exact acceptance matrix (7.3)–(7.5) and the R-129 shifted-Douglas anchor.

11. Devil’s-advocate review and corrected draft defects

1. **Objection: the finite matrix fixture constructs the production owner-complete response. UPHOLD.** It proves only the conditional factorization (3.5); production owner completeness and uniform boundedness remain open.
2. **Objection: the first Fourier fixture used shell labels as Fourier frequencies. VALID WITH MITIGATION.** The released fixture uses the actual modes $2^r, 2^m$ and shift $2^m - 2^r$.
3. **Objection: the selected coefficient $1/2$ was called the multiplier supremum. VALID WITH MITIGATION.** The released text separates the shell coefficient $1/2$ from $\|Q\|_\infty = 1$.
4. **Objection: the Ξ lower bound was stated for all $\lambda \geq 0$. VALID WITH MITIGATION.** Equation (8.7) is restricted to $0 \leq \lambda \leq 1$, and production has $\lambda \leq 5/9$.
5. **Objection: the A8 rounded c_{sym} was treated as exact. VALID WITH MITIGATION.** The released programs recompute (7.7) from A1 and use A8 only as a decimal regression.
6. **Objection: derived A1 coefficients and R-130 frame constants were hardcoded. VALID WITH MITIGATION.** The released programs derive $a, b, c, \alpha, c_0, c_1$ from A1 and import hash-pinned L_6, H_6 from R-130.

7. **Objection: diagonal Gram regularity controls the conditional mean square. UPHELD against the inference.** Equations (5.2)–(5.5) give an exact counterfixture.
8. **Objection: a bounded complete coefficient already has dyadic decay. UPHELD against the inference.** Equation (6.2) saturates a distant shell with coefficient $1/2$.
9. **Objection: (4.5) is the missing complete production response. UPHELD.** It is only the deterministic current-square component.
10. **Objection: the local Cartan ratio proves global acceptance. UPHELD.** It omits the complete mixed, far, same-channel, and low owners.
11. **Objection: the Ξ radial coefficient bound controls the full tangent norm after only the natural common-phase quotient. UPHELD.** The explicit phase-invisible fixture after (8.7) has $a = s = h = 0$ but nonzero tangent norm, and the sharp radial coefficient in (8.8) also vanishes as λ^2 .
12. **Objection: any fixed finite-moment heat restores a uniform gap. UPHELD against the claim.** Dominated convergence proves the fixed-law failure in Section 9.
13. **Objection: the sextic ray Hessian closes the complement. UPHELD as an objection to closure.** It is promising but has not been coupled to the complete response, low block, and anchor.
14. **Objection: the source and sextic may be reused after force completion. UPHELD against reuse.** Their coefficients are imported from the one-use R-103 budget, and the source Hessian stays outside the physical kernel.
15. **Objection: R-131 closes A13 or Sector A. DISMISSED.** The listed production constants, matching energy, anchor, and downstream Nelson chain remain open.

12. Executed evidence and reproduction

The primary program uses SymPy exact algebra. It reads A1 and the accepted R-103, R-124, and R-130 evidence, derives every new constant from those sources, checks the conditional response identity, exact counterfixtures, acceptance polynomials, Ξ spectrum, and heat boundary, and writes its JSON atomically. The independent program uses only the Python standard library; it does not import or read the R-131 primary output. It uses rational matrix and rotation algebra, a Laurent-mode fixture, an exact bounded rational state multiplier, stable eigenvalues, and a 64-atom heat covariance audit with a collapsed four-value response.

Run

```
E:\Dev\TECT.venv\Scripts\python.exe \
  codes/foundations/a13_classii_owner_complete_physical_response_mixed_gram_shell_boundary.py

E:\Dev\TECT.venv\Scripts\python.exe \
  codes/foundations/a13_classii_owner_complete_physical_response_mixed_gram_shell_boundary_independent.py
```

The child outputs are

```
R-131 primary PASS: 69/69
R-131 independent PASS: 61/61
```

The integrated verifier reruns both children, checks cross-route constants, authority and file hashes, exact exploration and negative records, note/PDF pairing, PDF security and text, rendered-page identity, generated public surfaces, and the explicit false scope flags.

13. Result footer

Result ID: A13-CLASSII-OWNER-COMPLETE-PHYSICAL-RESPONSE-MIXED-GRAM-SHELL-BOUNDARY (R-131).

Precise statement: the action owner and one-use coefficients are fixed in Section 2; Section 3 proves the finite-cylinder response factorization conditional on an owner-complete physical form; Section 4 proves the deterministic current-square H2 component bound with coefficient $46959/(2000P)$; Sections 5 and 6 prove the diagonal-Gram and bounded-multiplier inference no-gos; Section 7 proves the conditional acceptance simplex; Section 8 proves the stratified Ξ radial coefficient bound and the natural phase-horizontal full-norm no-go; Section 9 proves fixed-heat non-uniformity and records the sextic complement.

Scope: fixed finite cutoff and positive floor; conditional owner-complete finite physical form for the factorization; deterministic current-square component for the H2 bound; method counterfixtures for mixed Gram and shell inference; production Ξ range $0 \leq \lambda \leq 5/9$.

Dependencies: A1, A8; R-103, R-120, R-121, R-123, R-124, R-128, R-129, and R-130.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_owner_complete_physical_response_mixed_gram_shell_boundary_verify.py

Expected output: R-131 integrated PASS; primary 69/69 and independent 61/61; manifest, PDF, exploration, negative-result, and generated-surface gates pass with deterministic integrated JSON.

Falsification gate: failure of the A1 completed-square coefficients; R-103 source/sextic one-use factors; R-130 L6/H6 import; conditional R*F response identity; deterministic H2 coefficient; rotation or dyadic Fourier counterfixture; bounded rational derivative fixture; acceptance polynomial; Ξ radial spectrum/domain certificate or common-phase full-norm counterfixture; 64-atom product-heat covariance and collapsed response; heat domination/asymptotic; any authority/hash, PDF, exploration, registry, or public-surface check.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no construction or uniform bound of the production owner-complete response, C_{mix} , C_{far} , c_{bal} , R_0 , S_0 , d , k , matching energy, absolute anchor, strict augmented gap, $\text{OVERLAP}_{\text{src}}$, Nelson, removal, interacting measure, Sector-A closure, or T5--T7 result follows.

Next required action: prove an owner-complete mixed-replica spatial multiplier theorem retaining trace, heat, forest, rational, Cartan, and low owners; combine it with a state/scale-adapted transverse source-sextic complement; compute the full acceptance matrix and apply the R-129 shifted-Douglas anchor without double spending.

Proof complete: false. Sector A closed: false.

No-overclaim footer. R-131 proves a conditional finite-cylinder factorization, one deterministic current-square H^2 component bound, four scoped no-gos, an exact conditional acceptance simplex, stratified Ξ radial-coefficient coercivity, and fixed-heat non-uniformity. It does not construct or uniformly bound the owner-complete production response; prove complete C_{mix} , C_{far} , c_{bal} , low constants, full-tangent Ξ coercivity, matching energy, or an anchor; close either A13 gate; establish $\text{OVERLAP}_{\text{src}}$, Nelson, removals, an interacting measure, or Sector A; or promote any claim to T5--T7.